Priyanshu Kumar

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Research Interests

Computational structural biology, AlphaFold-based protein structure prediction, machine learning for bioinformatics, physics-informed neural networks for biomolecular modeling, and development of context-aware protein refinement methodologies.

Education

Chandigarh University **Punjab, India**
Bachelor of Engineering in Biotechnology *Aug 2024 – Jul 2028*
CGPA: 9.2/10 (First Year) | **Department Rank:** 1/107 students
Relevant Coursework: Computational Biology, Molecular Genetics, Bioinformatics, Systems Biology

Research & Publications

bioRxiv | Under Review - The Journal of Supercomputing (Springer)
First Author Publication *2025*
Title: "AF2Rank Revisited: Reproducing AlphaFold-Based Structure Evaluation and a Hypothesis for Context-Aware Refinement (CAR-AF)"
Status: Preprint published (DOI: 10.1101/2025.04.30.651434), submitted to Springer journal (passed technical review)
• **Independent research:** Reproduced and extended AF2Rank methodology, processed 145,000+ protein decoy structures
• **Machine learning innovation:** The designed reverse classification model achieves 61.5% precision across 133 protein targets
• **Novel methodology:** Proposed Context-Aware Refinement (CAR-AF) framework to improve AlphaFold predictions
• **Complete implementation:** Deployed XGBoost pipeline with SHAP interpretability and reduction of dimensionality.

Technical Projects

AF2Rank Reproduction & Extension: • Implemented comprehensive AlphaFold evaluation pipeline utilizing JAX and TensorFlow frameworks
• Constructed feature engineering architecture incorporating 17 derived structural metrics
• Deployed optimized XGBoost classifier with Optuna-based hyperparameter optimization
• Developed reproducible computational notebooks for structural bioinformatics workflows

Physics-Informed Neural Networks for Protein Dynamics (Ongoing): • Collaborating with mathematics faculty on PINNs integration for structural biology applications
• Developing hybrid architecture combining differential equations with protein structure modeling
• Investigating deep learning approaches enhanced by physics-based constraints

AI-Enhanced E-Book Reader Application: • Developed full-stack EPUB/PDF reader with React frontend and Node.js backend architecture
• Implemented advanced features: text highlighting, annotation system, bookmarking, and library management
• Built responsive UI with dark/light mode toggle and customizable reading preferences
• Open-source project available on GitHub with comprehensive documentation and setup instructions

Technical Skills

Programming Languages: Python, C++, JavaScript, Bash, LaTeX

Machine Learning & AI: PyTorch, TensorFlow, XGBoost, scikit-learn, SHAP, Optuna

Structural Bioinformatics: AlphaFold2, ColabFold, Biopython, Rosetta Suite, PyMOL, ChimeraX

Data Science & Analytics: Pandas, NumPy, Matplotlib, Seaborn, PCA/t-SNE/UMAP, Statistical Analysis

Development & Tools: Git/GitHub, Google Colab, VSCode, Linux (WSL), Docker, High-Performance Computing

Achievements & Certifications

- **Tech Invent 2024:** Second Place - Developed IoT-based Water Quality detection system utilizing biosensor technologies
- **Professional Development:** Mathematics for Machine Learning, Deep Learning Specialization- In Progress

Research Vision

Developing innovative computational frameworks that integrate deep learning methodologies with physics-based modeling to advance protein structure prediction capabilities. The research focus includes the use of artificial intelligence for drug discovery, enzyme engineering, and the elucidation of protein-environment interactions. The long-term objective includes contributing to the challenges of CASP (Critical Assessment of Protein Structure Prediction) and translating computational innovations into therapeutic applications.